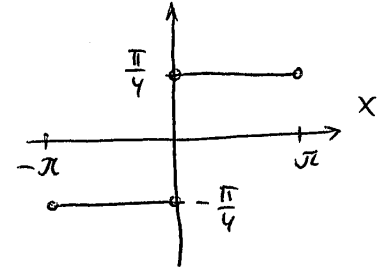


Fourier series.

SOLUTIONS TO EXAMPLES

1. (a) The function

$$f(x) = \begin{cases} -\frac{\pi}{4}, & -\pi < x < 0, \\ \frac{\pi}{4}, & 0 < x < \pi, \end{cases}$$



is odd. The product $f(x) \cos nx$ is an odd function, hence its integral over a symmetric interval is zero and $a_n = 0$.

On the other hand, the product $f(x) \sin nx$ is an even function. Hence, we have

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx \\ &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx \\ &= \frac{2}{\pi} \int_0^{\pi} \frac{\pi}{4} \sin nx \, dx \\ &= \frac{1}{2} \int_0^{\pi} \sin nx \, dx \\ &= \frac{1}{2} \frac{1}{n} [-\cos nx]_0^{\pi} \\ &= \frac{1}{2n} [-\cos n\pi - (-\cos 0)] \\ &= \frac{1 - (-1)^n}{2n}. \end{aligned}$$

The last formula above was obtained using $\cos 0 = 1$ and $\cos n\pi = (-1)^n$. It gives $b_n = 0$ for even n and $b_n = 1/n$ for odd n values ($n = 1, 3, \dots$). A generic odd number can be written as $n = 2m + 1$ with $m = 0, 1, 2, \dots$. Hence, the Fourier series for $f(x)$ is

$$\sum_{m=0}^{\infty} \frac{\sin(2m+1)x}{2m+1}. \quad (1)$$

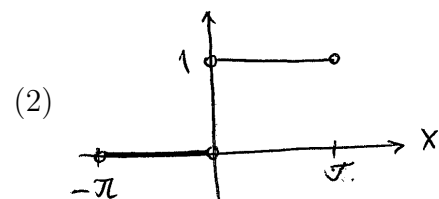
For $x = 0$, the series gives 0, which is $\frac{1}{2}[f(0-0) + f(0+0)] = \frac{1}{2} \left[-\frac{\pi}{4} + \frac{\pi}{4} \right] = 0$.

Note that for $x = \pi/2$, we have $\sin(2m+1)\frac{\pi}{2} = \sin\left(\frac{\pi}{2} + m\pi\right) = (-1)^m$, and the series gives

$$\sum_{m=0}^{\infty} \frac{(-1)^m}{2m+1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}.$$

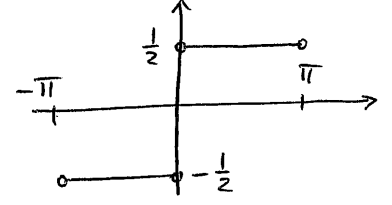
- (b) The function

$$f(x) = \begin{cases} 0, & -\pi < x < 0, \\ 1, & 0 < x < \pi, \end{cases}$$



is similar to that considered in part (a).

At $x = 0$ the function (2) has a discontinuity (jump), equal to 1, while $f(x)$ from part (a) has a jump equal to $\pi/2$. If we multiply that function by $2/\pi$, it will take the form shown on the right, i.e., taking the values $f(x) = -1/2$ for $-\pi < x < 0$ and $f(x) = 1/2$ for $0 < x < \pi$.



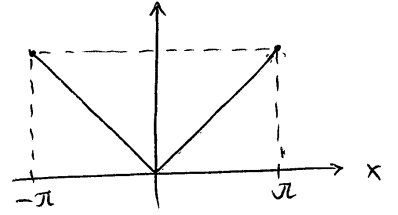
By adding $1/2$ to the function above gives $f(x)$ from Eq. (2). Hence, its Fourier series can be obtained by multiplying the series on the right-hand side of Eq. (1) by $2/\pi$ and adding $1/2$, which gives

$$\frac{1}{2} + \frac{2}{\pi} \sum_{m=0}^{\infty} \frac{\sin(2m+1)x}{2m+1}.$$

For $x = 0$ this series gives $1/2$, which is exactly $\frac{1}{2}[f(0-0) + f(0+0)] = \frac{1}{2}[0 + 1] = \frac{1}{2}$.

- (c) The function $f(x) = |x|$, $-\pi \leq x \leq \pi$ is even, hence only a_n are nonzero, and only a_0 and the cosine terms will be present. Since $f(x) \cos nx$ is even, we also have

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx.$$



For $n = 0$, we have

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x \, dx = \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} = \frac{2}{\pi} \frac{\pi^2}{2} = \pi.$$

For $n > 0$,

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx \\ &= \frac{2}{\pi} \frac{1}{n} \int_0^{\pi} x d(\sin nx) \\ &= \frac{2}{\pi n} \underbrace{[x \sin nx]_0^{\pi}}_{=0} - \frac{2}{\pi n} \int_0^{\pi} \sin nx \, dx \\ &= -\frac{2}{\pi n^2} [-\cos nx]_0^{\pi} = -\frac{2(1 - \cos n\pi)}{\pi n^2}. \end{aligned}$$

Since $\cos n\pi = (-1)^n$, we have

$$a_n = -\frac{2(1 - (-1)^n)}{\pi n^2} = \begin{cases} 0, & n \text{ even}, \\ -\frac{4}{\pi n^2}, & n \text{ odd}. \end{cases}$$

Using $2m+1$ for a generic odd number, we obtain the Fourier series for $f(x) = |x|$ as

$$\frac{\pi}{2} + \frac{4}{\pi} \sum_{m=0}^{\infty} \frac{\cos(2m+1)x}{(2m+1)^2}. \quad (3)$$

2. (a) The function $f(x) = \cos \alpha x$ is an even function, hence $b_n = 0$, and we find a_n as

$$a_n = \frac{2}{\pi} \int_0^{\pi} \cos \alpha x \cos nx \, dx.$$

For $n = 0$,

$$a_0 = \frac{2}{\pi} \int_0^\pi \cos \alpha x \, dx = \frac{2}{\pi} \frac{1}{\alpha} [\sin \alpha x]_0^\pi = \frac{2}{\pi \alpha} [\sin \alpha \pi - \sin 0],$$

so that

$$a_0 = \frac{2 \sin \alpha \pi}{\alpha \pi}.$$

For $n > 0$,

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^\pi \cos \alpha x \cos nx \, dx \\ &= \frac{2}{\pi} \int_0^\pi \frac{1}{2} [\cos(\alpha + n)x + \cos(\alpha - n)x] \, dx \\ &= \frac{1}{\pi} \left[\frac{1}{\alpha + n} \sin(\alpha + n)x + \frac{1}{\alpha - n} \sin(\alpha - n)x \right]_0^\pi \\ &= \frac{1}{\pi} \left[\frac{\sin(\alpha + n)\pi}{\alpha + n} + \frac{\sin(\alpha - n)\pi}{\alpha - n} \right] \\ &= \frac{1}{\pi} \left[\frac{(-1)^n \sin \alpha \pi}{\alpha + n} + \frac{(-1)^n \sin \alpha \pi}{\alpha - n} \right] \\ &= (-1)^n \frac{\sin \alpha \pi}{\pi} \left(\frac{1}{\alpha + n} + \frac{1}{\alpha - n} \right). \end{aligned}$$

In the above, we used the identity¹

$$\sin(\alpha \pm n)\pi = \sin(\alpha\pi \pm n\pi) = \sin \alpha\pi \cos n\pi \pm \cos \alpha\pi \sin n\pi = (-1)^n \sin \alpha\pi.$$

Hence, the Fourier series is

$$\cos \alpha x = \frac{\sin \alpha \pi}{\alpha \pi} + \frac{\sin \alpha \pi}{\pi} \sum_{n=1}^{\infty} (-1)^n \left(\frac{1}{\alpha + n} + \frac{1}{\alpha - n} \right) \cos nx, \quad (4)$$

or

$$\cos \alpha x = \frac{\sin \alpha \pi}{\pi} \left[\frac{1}{\alpha} + \sum_{n=1}^{\infty} (-1)^n \left(\frac{1}{\alpha + n} + \frac{1}{\alpha - n} \right) \cos nx \right]. \quad (5)$$

(b) Setting $x = 0$ in Eq. (5), we have

$$1 = \frac{\sin \alpha \pi}{\pi} \left[\frac{1}{\alpha} + \sum_{n=1}^{\infty} (-1)^n \left(\frac{1}{\alpha + n} + \frac{1}{\alpha - n} \right) \right],$$

or

$$\frac{\pi}{\sin \alpha \pi} = \frac{1}{\alpha} + \sum_{n=1}^{\infty} (-1)^n \left(\frac{1}{\alpha + n} + \frac{1}{\alpha - n} \right).$$

(c) Setting $x = \pi$ in Eq. (5) gives

$$\cos \alpha \pi = \frac{\sin \alpha \pi}{\pi} \left[\frac{1}{\alpha} + \sum_{n=1}^{\infty} (-1)^n \left(\frac{1}{\alpha + n} + \frac{1}{\alpha - n} \right) \underbrace{\cos n\pi}_{(-1)^n} \right],$$

¹This identity can be “seen” easily from the picture of a unit circle for $\sin \theta$ and $\cos \theta$. Adding $n\pi$ to the angle θ either keeps the point on the unit circle in the same place (even n), or puts it diagonally opposite (odd n). In the first case $\sin(\theta + n\pi)$ and $\cos(\theta + n\pi)$ are the same as $\sin \theta$ and $\cos \theta$; in the second case, $\sin(\theta + n\pi) = -\sin \theta$ and $\cos(\theta + n\pi) = -\cos \theta$.

or

$$\frac{\pi \cos \alpha \pi}{\sin \alpha \pi} = \frac{1}{\alpha} + \sum_{n=1}^{\infty} \left(\frac{1}{\alpha + n} + \frac{1}{\alpha - n} \right),$$

which can be written as

$$\pi \cot \alpha \pi = \frac{1}{\alpha} + \sum_{n=1}^{\infty} \left(\frac{1}{\alpha + n} + \frac{1}{\alpha - n} \right). \quad (6)$$

Note that this formula immediately shows that $\cot \theta \rightarrow \infty$ for $\theta \rightarrow m\pi$ ($m \in \mathbb{Z}$), as can be seen from the right-hand side in the case $\alpha \rightarrow m$, when one of the denominators on the right-hand side tends to zero.

Taking $\alpha = \frac{1}{4}$ in Eq. (6) gives

$$\pi \cot \frac{\pi}{4} = 4 + \sum_{n=1}^{\infty} \left(\frac{1}{\frac{1}{4} + n} + \frac{1}{\frac{1}{4} - n} \right).$$

Since $\cot \frac{\pi}{4} = 1$, we have

$$\begin{aligned} \pi &= 4 + \sum_{n=1}^{\infty} \frac{\frac{1}{4} - n + \frac{1}{4} + n}{\left(\frac{1}{4} + n\right) \left(\frac{1}{4} - n\right)}, \\ \pi &= 4 + \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{\frac{1}{16} - n^2}, \end{aligned}$$

which finally gives

$$\pi = 4 - \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n^2 - \frac{1}{16}}. \quad (7)$$

This equation can be used to calculate π numerically by computing the sum on the right-hand side. Summing the first 100 and 1000 terms in the series gives the following answers:

$$\begin{aligned} 4 - \frac{1}{2} \sum_{n=1}^{100} \frac{1}{n^2 - \frac{1}{16}} &\approx 3.14657, \\ 4 - \frac{1}{2} \sum_{n=1}^{1000} \frac{1}{n^2 - \frac{1}{16}} &\approx 3.14209. \end{aligned}$$

The error in the approximate value of π obtained by adding the terms up to $n = N$ is

$$-\frac{1}{2} \sum_{n=N+1}^{\infty} \frac{1}{n^2 - \frac{1}{16}}.$$

This sum can be estimated quickly by the integral

$$-\frac{1}{2} \int_{N+1}^{\infty} \frac{1}{n^2 - \frac{1}{16}} dn \simeq -\frac{1}{2} \int_{N+1}^{\infty} \frac{dn}{n^2} = -\frac{1}{2} \left[-\frac{1}{n} \right]_{N+1}^{\infty} = -\frac{1}{2(N+1)}.$$

Adding this error to the numerical estimates of π obtained above gives

$$3.14657 - \frac{1}{202} \approx 3.14162, \quad 3.14209 - \frac{1}{2002} \approx 3.14159,$$

whether the latter is accurate to 5 decimal places.